

理论力学作业

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2021 年 1 月 1 日

第六章

6.1 $L'(\dot{p}, p, t)$ 的全微分为:

$$dL' = \frac{\partial L'}{\partial \dot{p}} d\dot{p} + \frac{\partial L'}{\partial p} dp + \frac{\partial L'}{\partial t} dt$$

同时也可以表示为:

$$\begin{aligned} dL' &= dH + \dot{p}dq + qd\dot{p} \\ &= \left(\dot{q}dp - \dot{p}dq + \frac{\partial H}{\partial t} dt \right) + \dot{p}dq + qd\dot{p} \\ &= qd\dot{p} + \dot{q}dp + \frac{\partial H}{\partial t} dt \end{aligned}$$

比较得:

$$\frac{\partial L'}{\partial \dot{p}} = q, \quad \frac{\partial L'}{\partial p} = \dot{q} = \frac{dq}{dt}, \quad \frac{\partial L'}{\partial t} dt = \frac{\partial H}{\partial t} dt$$

则显然有:

$$\frac{d}{dt} \left(\frac{\partial L'}{\partial \dot{p}} \right) - \frac{\partial L'}{\partial p} = 0$$

6.2 质点的 Lagrange 量为:

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}(k_1 + k_2)x^2$$

质点的 Hamilton 量为:

$$H = \frac{p^2}{2m} + \frac{1}{2}(k_1 + k_2)x^2$$

正则方程为:

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m} \quad (1)$$

$$\dot{p} = m\ddot{x} = -\frac{\partial H}{\partial x} = -(k_1 + k_2)x \quad (2)$$

方程 (1) 显然成立, 而由方程 (2) 得:

$$\ddot{x} + \frac{k_1 + k_2}{m}x = 0$$

解得：

$$x = A \cos(\omega t + \phi), \quad \omega = \sqrt{\frac{k_1 + k_2}{m}}, \quad A < \frac{a}{2}$$

其中振幅 A 与相位 ϕ 由初始条件决定。

6.3 单摆坐标为：

$$\mathbf{r} = (x + l \sin \theta) \hat{\mathbf{x}} + (ax^2 - l \cos \theta) \hat{\mathbf{z}}$$

则单摆速度为：

$$\mathbf{v} = (\dot{x} + \dot{\theta} l \cos \theta) \hat{\mathbf{x}} + (2ax\dot{x} + \dot{\theta} l \sin \theta) \hat{\mathbf{y}}$$

系统的 Lagrange 量为：

$$\begin{aligned} L &= T - V \\ &= \frac{1}{2}m(\dot{x} + \dot{\theta} l \cos \theta)^2 - \frac{1}{2}m(2ax\dot{x} + \dot{\theta} l \sin \theta)^2 + mg(ax^2 - l \cos \theta) \\ &= \frac{1}{2}m(1 + 4a^2x^2)\dot{x}^2 - \frac{1}{2}ml^2\dot{\theta}^2 + ml(\cos \theta + 2ax \sin \theta)\dot{x}\dot{\theta} - mg(ax^2 - l \cos \theta) \end{aligned}$$

与 x 、 θ 共轭的广义动量分别为：

$$\begin{aligned} p_x &= \frac{\partial L}{\partial \dot{x}} = m(1 + 4a^2x^2)\dot{x} + ml(\cos \theta + 2ax \sin \theta)\dot{\theta} = A\dot{x} + B\dot{\theta} \\ p_\theta &= \frac{\partial L}{\partial \dot{\theta}} = ml^2\dot{\theta} + ml(\cos \theta + 2ax \sin \theta)\dot{x} = C\dot{x} + D\dot{\theta} \end{aligned}$$

其中：

$$A = m(1 + 4a^2x^2), \quad B = C = ml(\cos \theta + 2ax \sin \theta), \quad D = ml^2$$

解得：

$$\dot{x} = \frac{Dp_x - Bp_\theta}{AD - BC}, \quad \dot{\theta} = \frac{Ap_\theta - Cp_x}{AD - BC}$$

则系统的 Hamilton 量为：

$$\begin{aligned} H &= T + V \\ &= \frac{1}{2}m(\dot{x} + \dot{\theta} l \cos \theta)^2 + \frac{1}{2}m(2ax\dot{x} + \dot{\theta} l \sin \theta)^2 + mg(ax^2 - l \cos \theta) \\ &= \frac{1}{2}m(1 + 4a^2x^2)\dot{x}^2 - \frac{1}{2}ml^2\dot{\theta}^2 + ml(\cos \theta + 2ax \sin \theta)\dot{x}\dot{\theta} + mg(ax^2 - l \cos \theta) \\ &= \frac{A}{2} \cdot \left(\frac{Dp_x - Bp_\theta}{AD - BC} \right)^2 + \frac{D}{2} \cdot \left(\frac{Ap_\theta - Cp_x}{AD - BC} \right)^2 + \frac{B(Dp_x - Bp_\theta)(Ap_\theta - Cp_x)}{(AD - BC)^2} + mg(ax^2 - l \cos \theta) \\ &= \frac{Dp_x^2 + Ap_\theta^2 - (B + C)p_x p_\theta}{2(AD - BC)} \\ &= \frac{[p_\theta, p_x] \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} p_\theta \\ p_x \end{bmatrix}}{2 \begin{vmatrix} A & B \\ C & D \end{vmatrix}} + mg(ax^2 - l \cos \theta) \end{aligned}$$

其中：

$$A = m(1 + 4a^2x^2), \quad B = C = ml(\cos \theta + 2ax \sin \theta), \quad D = ml^2$$

6.5.(1) 非相对论情况下, Lagrange 量为:

$$L = \frac{1}{2}m\mathbf{v}^2 + e\mathbf{v} \cdot \mathbf{A} - eV(r)$$

正则动量为:

$$\mathbf{P} = \frac{\partial L}{\partial \mathbf{v}} = m\mathbf{v} + e\mathbf{A}$$

Hamilton 量为:

$$L = \frac{1}{2m}(\mathbf{P} - e\mathbf{A})^2 + eV(r) = \frac{1}{2m} \left(\mathbf{P} - \frac{e}{2}\mathbf{B}_0 \times \mathbf{r} \right)^2 + eV(r)$$

6.5.(2) 在极坐标系下,

$$\mathbf{A} = \frac{1}{2}\mathbf{B}_0 \times \mathbf{r} = \frac{1}{2}(B_0 \hat{\mathbf{z}}) \times (r\hat{\mathbf{r}}) = \frac{1}{2}B_0 r \hat{\boldsymbol{\theta}}$$

则 Lagrange 量为:

$$\begin{aligned} L &= \frac{1}{2}m\mathbf{v}^2 + e\mathbf{v} \cdot \mathbf{A} - eV(r) \\ &= \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + e(\dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}}) \cdot \left(\frac{1}{2}B_0 r \hat{\boldsymbol{\theta}} \right) - eV(r) \\ &= \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{1}{2}eB_0 r^2 \dot{\theta} - eV(r) \end{aligned}$$

换系之后, Lagrange 量变为:

$$\begin{aligned} L &= \frac{1}{2}m \left[\dot{r}^2 + r^2 \left(\dot{\theta} - \frac{eB_0}{2m} \right)^2 \right] + \frac{1}{2}eB_0 r^2 \left(\dot{\theta} - \frac{eB_0}{2m} \right) - eV(r) \\ &= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 - \frac{(eB_0 r)^2}{8m} - eV(r) \end{aligned}$$

正则动量为:

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}, \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}$$

Hamilton 量为:

$$\begin{aligned} H &= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 - \frac{(eB_0 r)^2}{8m} + eV(r) \\ &= \frac{1}{2m}(p_r^2 + p_\theta^2) - \frac{(eB_0 r)^2}{8m} + eV(r) \\ &= \frac{1}{2m}(\mathbf{P}^2 - e\mathbf{A}^2) + eV(r) \end{aligned}$$

6.6.(1) 在极坐标中, 系统的 Lagrange 量为:

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{GMm}{r}$$

则正则动量为:

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}, \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}$$

Laplace-Runge-Lenz 矢量为:

$$\begin{aligned}
 \mathbf{e} &= \frac{1}{GMm^2}(\mathbf{p} \times \mathbf{J}) - \frac{\mathbf{r}}{r} \\
 &= \frac{1}{GMm^2}(m\dot{\mathbf{r}} \times m\mathbf{r} + mr\dot{\boldsymbol{\theta}}) \times (mr^2\dot{\boldsymbol{\theta}}) - \hat{\mathbf{r}} \\
 &= \frac{1}{GM}(r^3\dot{\boldsymbol{\theta}}^2 \hat{\mathbf{r}} - r^2\dot{\mathbf{r}}\dot{\boldsymbol{\theta}}) - \hat{\mathbf{r}} \\
 &= \frac{1}{GMm^2} \left(\frac{p_\theta^2}{r} \hat{\mathbf{r}} - p_r p_\theta \hat{\boldsymbol{\theta}} \right) - \hat{\mathbf{r}}
 \end{aligned}$$

Hamilton 量为:

$$H = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{GMm}{r} = \frac{1}{2m}(p_r^2 + \frac{p_\theta^2}{r^2}) - \frac{GMm}{r}$$

由 $\frac{\partial \hat{\mathbf{r}}}{\partial r} = 0$ 、 $\frac{\partial \hat{\mathbf{r}}}{\partial \theta} = \hat{\boldsymbol{\theta}}$ 、 $\frac{\partial \hat{\boldsymbol{\theta}}}{\partial r} = 0$ 、 $\frac{\partial \hat{\boldsymbol{\theta}}}{\partial \theta} = -\hat{\mathbf{r}}$ 得:

$$\frac{\partial \mathbf{e}}{\partial r} = -\frac{1}{GMm^2} \left(\frac{p_\theta^2}{r^2} \hat{\mathbf{r}} \right), \quad \frac{\partial \mathbf{e}}{\partial \theta} = \frac{1}{GMm^2} \left(\frac{p_\theta^2}{r} \hat{\boldsymbol{\theta}} + p_r p_\theta \hat{\mathbf{r}} \right) - \hat{\boldsymbol{\theta}}$$

$$\begin{aligned}
 \frac{\partial \mathbf{e}}{\partial p_r} &= -\frac{p_\theta \hat{\boldsymbol{\theta}}}{GMm^2}, & \frac{\partial \mathbf{e}}{\partial p_\theta} &= -\frac{p_r \hat{\boldsymbol{\theta}}}{GMm^2} \\
 \frac{\partial H}{\partial r} &= -\frac{p_\theta^2}{mr^3} + \frac{GMm}{r^2}, & \frac{\partial H}{\partial \theta} &= 0 \\
 \frac{\partial H}{\partial p_r} &= \frac{p_r}{m}, & \frac{\partial H}{\partial p_\theta} &= \frac{p_\theta}{mr^2}
 \end{aligned}$$

则 Poisson 括号为:

$$\begin{aligned}
 [\mathbf{e}, H] &= \left(\frac{\partial \mathbf{e}}{\partial r} \cdot \frac{\partial H}{\partial p_r} + \frac{\partial \mathbf{e}}{\partial \theta} \cdot \frac{\partial H}{\partial p_\theta} \right) - \left(\frac{\partial \mathbf{e}}{\partial r} \cdot \frac{\partial H}{\partial r} + \frac{\partial \mathbf{e}}{\partial p_\theta} \cdot \frac{\partial H}{\partial \theta} \right) \\
 &= -\frac{1}{GMm^2} \left(\frac{p_\theta^2}{r^2} \hat{\mathbf{r}} \right) \cdot \frac{p_r}{m} + \left[\frac{1}{GMm^2} \left(\frac{p_\theta^2}{r} \hat{\boldsymbol{\theta}} + p_r p_\theta \hat{\mathbf{r}} \right) - \hat{\boldsymbol{\theta}} \right] \cdot \frac{p_\theta}{mr^2} \\
 &\quad + \frac{p_\theta \hat{\boldsymbol{\theta}}}{GMm^2} \cdot \left(-\frac{p_\theta^2}{mr^3} + \frac{GMm}{r^2} \right) \\
 &= -\frac{1}{GMm^3} \cdot \frac{p_\theta p_r}{r^2} \hat{\mathbf{r}} + \frac{1}{GMm^3} \cdot \frac{p_\theta^3}{r^3} \hat{\boldsymbol{\theta}} + \frac{1}{GMm^3} \cdot \frac{p_\theta p_r}{r^2} \hat{\mathbf{r}} - \frac{p_\theta}{mr^2} \hat{\boldsymbol{\theta}} + \frac{p_\theta}{mr^2} \hat{\boldsymbol{\theta}} - \frac{1}{GMm^3} \cdot \frac{p_\theta^3}{r^3} \hat{\boldsymbol{\theta}} \\
 &= 0
 \end{aligned}$$

则 $\mathbf{e}$ 守恒, 为常矢量。

6.6.(2) 取 $\mathbf{e}$ 为极轴方向, 则 $\mathbf{r} \cdot \mathbf{e} = re \cos \theta$

同时又有:

$$\begin{aligned}
 \mathbf{r} \cdot \mathbf{e} &= \mathbf{r} \cdot \left[\frac{1}{GMm^2}(\mathbf{p} \times \mathbf{J}) - \frac{\mathbf{r}}{r} \right] \\
 &= \frac{1}{GMm^2} [\mathbf{r} \cdot (\mathbf{p} \times \mathbf{J})] - r \\
 &= \frac{1}{GMm^2} [\mathbf{J} \cdot (\mathbf{r} \times \mathbf{p})] - r \\
 &= \frac{J^2}{GMm^2} - r
 \end{aligned}$$

即：

$$re \cos \theta = \frac{J^2}{GMm^2} - r$$

化简得轨道方程：

$$r = \frac{J^2}{GMm^2} \cdot \frac{1}{1 + e \cos \theta}$$

这是极坐标系下，极轴沿长轴，偏心率为 e 的椭圆方程。

说明矢量 $\mathbf{e}$ 取由力心到近日点的方向，大小等于偏心率 e 。

7.3.(1) 由

$$Q = \ln \left(\frac{\sin p}{q} \right), \quad P = q \cot p$$

得：

$$q = \frac{\sqrt{1 - P^2 e^{2Q}}}{e^Q}, \quad p = \arccos P e^Q$$

于是有：

$$\begin{aligned} \frac{\partial p}{\partial P} &= \frac{-e^Q}{\sqrt{1 - P^2 e^{2Q}}}, & \frac{\partial p}{\partial Q} &= \frac{-P e^Q}{\sqrt{1 - P^2 e^{2Q}}} \\ \frac{\partial q}{\partial P} &= \frac{-P e^Q}{\sqrt{1 - P^2 e^{2Q}}}, & \frac{\partial q}{\partial Q} &= -\frac{\sqrt{1 - P^2 e^{2Q}}}{e^Q} - \frac{P^2 e^Q}{\sqrt{1 - P^2 e^{2Q}}} \end{aligned}$$

则：

$$\begin{aligned} [q, p]_{Q, P} &= \frac{\partial q}{\partial Q} \cdot \frac{\partial p}{\partial P} - \frac{\partial q}{\partial P} \cdot \frac{\partial p}{\partial Q} \\ &= \left(-\frac{\sqrt{1 - P^2 e^{2Q}}}{e^Q} - \frac{P^2 e^Q}{\sqrt{1 - P^2 e^{2Q}}} \right) \cdot \left(\frac{-e^Q}{\sqrt{1 - P^2 e^{2Q}}} \right) - \\ &\quad \left(\frac{-P e^Q}{\sqrt{1 - P^2 e^{2Q}}} \right) \cdot \left(\frac{-P e^Q}{\sqrt{1 - P^2 e^{2Q}}} \right) \\ &= 1 + \frac{P^2 e^{2Q}}{1 - P^2 e^{2Q}} - \frac{P^2 e^{2Q}}{1 - P^2 e^{2Q}} \\ &= 1 \end{aligned}$$

$$[q, q]_{Q, P} = \frac{\partial q}{\partial Q} \cdot \frac{\partial q}{\partial P} - \frac{\partial q}{\partial P} \cdot \frac{\partial q}{\partial Q} = 0$$

$$[p, p]_{Q, P} = \frac{\partial p}{\partial Q} \cdot \frac{\partial p}{\partial P} - \frac{\partial p}{\partial P} \cdot \frac{\partial p}{\partial Q} = 0$$

即该变换保持最基本的 Poisson 括号不变，则它不改变任意两个力学量 f 、 g 之间的 Poisson 括号。

$$[f, g]_{p, q} = [f, g]_{P, Q}$$

故该变换为正则变换。

7.3.(2) 由

$$q = \sqrt{\frac{2Q}{k}} \cos P, \quad p = \sqrt{2Qk} \sin P$$

得：

$$Q = \frac{1}{2} \left(q^2 k + \frac{p^2}{k} \right), \quad P = \arctan \frac{p}{kq}$$

于是有：

$$\begin{aligned}\frac{\partial P}{\partial p} &= \frac{qk}{q^2k^2 + p^2}, & \frac{\partial P}{\partial q} &= -\frac{pk}{q^2k^2 + p^2} \\ \frac{\partial Q}{\partial p} &= \frac{p}{k}, & \frac{\partial Q}{\partial q} &= qk\end{aligned}$$

则：

$$\begin{aligned}[q, p]_{Q, P} &= \frac{\partial q}{\partial Q} \cdot \frac{\partial p}{\partial P} - \frac{\partial q}{\partial P} \cdot \frac{\partial p}{\partial Q} \\ &= qk \cdot \frac{qk}{q^2k^2 + p^2} - pk \cdot \left(-\frac{pk}{q^2k^2 + p^2} \right) \\ &= \frac{q^2k^2 + p^2}{q^2k^2 + p^2} \\ &= 1\end{aligned}$$

$$[q, q]_{Q, P} = \frac{\partial q}{\partial Q} \cdot \frac{\partial q}{\partial P} - \frac{\partial q}{\partial P} \cdot \frac{\partial q}{\partial Q} = 0$$

$$[p, p]_{Q, P} = \frac{\partial p}{\partial Q} \cdot \frac{\partial p}{\partial P} - \frac{\partial p}{\partial P} \cdot \frac{\partial p}{\partial Q} = 0$$

即该变换保持最基本的 Poisson 括号不变，则它不改变任意两个力学量 f 、 g 之间的 Poisson 括号。

$$[f, g]_{p, q} = [f, g]_{P, Q}$$

故该变换为正则变换。

3.(a) 由 $\frac{\partial S}{\partial q_i} = p_i$ 及 $\frac{\partial S}{\partial t} = -H$ 得：

$$\frac{\partial S}{\partial t} + \frac{1}{2m} \left(\frac{\partial S}{\partial \mathbf{x}} \right)^2 + \frac{1}{2} m \omega^2 \mathbf{x}^2$$

此即为作用量函数所满足的 Hamilton-Jacobi 方程。

3.(b) 令 $S(x_1, x_2, t) = T(t) + S_1(x_1) + S_2(x_2)$ ，分离变量得：

$$\frac{dT}{dt} + \frac{1}{2m} \left[\left(\frac{dS_1}{dx_1} \right)^2 + \left(\frac{dS_2}{dx_2} \right)^2 \right] + \frac{1}{2} m \omega^2 (x_1^2 + x_2^2) = 0$$

引入常数 σ 和 τ ，可以表达为：

$$\begin{cases} \frac{dT}{dt} + \sigma + \tau = 0 \\ \frac{1}{2m} \left(\frac{dS_1}{dx_1} \right)^2 + \frac{1}{2} m \omega^2 x_1^2 - \sigma = 0 \\ \frac{1}{2m} \left(\frac{dS_2}{dx_2} \right)^2 + \frac{1}{2} m \omega^2 x_2^2 - \tau = 0 \end{cases}$$

3.(c) 由题意, 令 $\sigma = Q_1$ 、 $\tau = Q_2$, 积分得:

$$\begin{aligned} T &= -(Q_1 + Q_2)t \\ S_1 &= \int_0^{x_1} \sqrt{2mQ_1 - m^2\omega^2\xi_1^2} d\xi_1 \\ S_2 &= \int_0^{x_2} \sqrt{2mQ_2 - m^2\omega^2\xi_2^2} d\xi_2 \end{aligned}$$

则 $S(x, y, t)$ 的完全解为:

$$S = \int_0^{x_1} \sqrt{2mQ_1 - m^2\omega^2\xi_1^2} d\xi_1 + \int_0^{x_2} \sqrt{2mQ_2 - m^2\omega^2\xi_2^2} d\xi_2 - (Q_1 + Q_2)t$$

3.(d) 新的广义动量 P_i 定义为:

$$P_i = -\frac{\partial S}{\partial Q_i} = -\int_0^{x_i} \frac{m}{\sqrt{2mQ_i - m^2\omega^2\xi_i^2}} d\xi_i + t$$

利用 (6.182) 式积分得:

$$P_i = -\frac{1}{\omega} \arccos\left(\sqrt{\frac{m\omega^2}{2Q_i}} x_i\right) + t$$

则:

$$x_i = \sqrt{\frac{2Q_i}{m\omega^2}} \cos(-\omega t + \omega P_i) = \sqrt{\frac{2Q_i}{m\omega^2}} \cos(\omega t - \omega P_i)$$

可以看出 P_i 与不同方向上振动的相位 $\Delta\phi_i = -\omega P_i$ 相关联。

3.(e) 由于粒子轨迹 $\mathbf{x}(t)$ 由两个方向的同频率简谐振动叠加而成, 且一般振幅、相位均不同, 说明粒子轨道是闭合的椭圆。

3.(f) 以下过程中令 $\phi_i = \omega t - \omega P_i$ 。

注意到:

$$I_i = \frac{1}{2\pi} \oint p_i dx_i = \frac{1}{2\pi} \int_0^T p_i \frac{dx_i}{dt} dt$$

其中周期 $T = \frac{2\pi}{\omega}$ 。

由 $x_i = \sqrt{\frac{2Q_i}{m\omega^2}} \cos \phi$ 得:

$$\frac{dx_i}{dt} = \omega \sqrt{\frac{2Q_i}{m\omega^2}} \sin \phi$$

又:

$$p_i = \frac{\partial S}{\partial x_i} = \sqrt{2mQ_i - m^2\omega^2 x_i^2} = \sqrt{2mQ_i} \sin \phi$$

则绝热不变量 I_i 为:

$$I_i = \frac{1}{2\pi} \int_0^T p_i \frac{dx_i}{dt} dt$$

$$\begin{aligned}
&= \frac{1}{2\pi} \int_0^T \omega \sqrt{\frac{2Q_i}{m\omega^2}} \sin \phi \cdot \sqrt{2mQ_i} \sin \phi \, dt \\
&= \frac{Q_i}{\pi\omega} \int_0^{2\pi} \sin^2 \phi \, d\phi \\
&= \frac{Q_i}{\omega}
\end{aligned}$$

由于总能量:

$$E = H = Q_1 + Q_2 = \omega(I_1 + I_2)$$

可见总能量仅仅依赖于它们特殊的线性组合。

各个 ω_i 为:

$$\omega_i = \frac{\partial E}{\partial I_i} = \omega$$

3.(g) 由 $\frac{\partial S}{\partial q_i} = p_i$ 及 $\frac{\partial S}{\partial t} = -H$ 得:

$$\frac{\partial S}{\partial t} + \frac{1}{2m} \left[\left(\frac{\partial S}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial S}{\partial \theta} \right)^2 \right] + \frac{1}{2} \omega^2 r^2$$

此即为作用量函数所满足的 Hamilton-Jacobi 方程。

令 $S(x_1, x_2, t) = T(t) + \Theta(\theta) + R(r)$, 分离变量得:

$$\frac{dT}{dt} + \frac{1}{2m} \left[\left(\frac{dR}{dr} \right)^2 + \frac{1}{r^2} \left(\frac{d\Theta}{d\theta} \right)^2 \right] + \frac{1}{2} m \omega^2 r^2 = 0$$

引入常数 λ 和 κ , 可以表达为:

$$\begin{cases} \frac{dT}{dt} + \lambda = 0 \\ \frac{1}{2m} \left(\frac{d\Theta}{d\theta} \right)^2 - \kappa = 0 \\ \frac{1}{2m} \left(\frac{dR}{dr} \right)^2 + \frac{1}{2} \omega^2 r^2 - \lambda + \frac{\kappa}{r^2} = 0 \end{cases}$$

积分得:

$$S = \int_0^r \sqrt{2m\lambda - \frac{2m\kappa}{\rho^2} - m^2\omega^2\rho^2} \, d\rho + \sqrt{2m\kappa}\theta - \lambda t$$

3.(h) 考虑与 r 及 θ 相关的广义动量:

$$\begin{aligned}
P_\kappa &= -\frac{\partial S}{\partial \kappa} \\
&= -\int_0^r \frac{m}{\rho^2 \sqrt{2m\lambda - \frac{2m\kappa}{\rho^2} - m^2\omega^2\rho^2}} \, d\rho + \sqrt{\frac{m}{2\kappa}} \theta
\end{aligned}$$

换元, 令 $u = \frac{1}{\rho^2}$, 并利用 (6.182) 式:

$$\begin{aligned}
 P_\kappa &= - \int_0^r \frac{m}{\rho^2 \sqrt{2m\lambda - \frac{2m\kappa}{\rho^2} - m^2\omega^2\rho^2}} d\rho + \sqrt{\frac{m}{2\kappa}}\theta \\
 &= - \int_0^r \frac{mu}{\sqrt{2m\lambda - 2m\kappa u - \frac{m^2\omega^2}{u}}} \left(-\frac{1}{2u\sqrt{u}}du\right) + \sqrt{\frac{m}{2\kappa}}\theta \\
 &= \sqrt{\frac{m}{8\kappa}} \int_0^r \frac{1}{\sqrt{-\frac{1}{2m\kappa}\omega^2 + \frac{\lambda}{m\kappa}u - u^2}} du + \sqrt{\frac{m}{2\kappa}}\theta \\
 &= \sqrt{\frac{m}{8\kappa}} \arccos\left(\frac{\lambda - 2m\kappa u}{\sqrt{\lambda^2 - 2\kappa\omega^2}}\right) + \sqrt{\frac{m}{2\kappa}}\theta
 \end{aligned}$$

代入回 $u = \frac{1}{r^2}$, 并整理得:

$$r^2 = \frac{2m\kappa}{(\lambda + \sqrt{\lambda^2 - 2\kappa\omega^2}) - 2\sqrt{\lambda^2 - 2\kappa\omega^2} \cos^2(\theta - \Delta\theta)}, \quad \Delta\theta = p_\kappa \sqrt{\frac{2\kappa}{m}}$$

即轨道方程具有以下形式:

$$r^2 = \frac{C_1}{C_2 - C_3 \cos^2 \theta}$$

代入 $r^2 = x^2 + y^2$ 及 $\cos \theta = \frac{x}{\sqrt{x^2 + y^2}}$ 可以化为:

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, \quad A = \sqrt{\frac{C_1}{C_2 - C_3}}, \quad B = \sqrt{\frac{C_1}{C_3}}$$

其中要求 $C_2 - C_3 > 0$, 而本题中

$$C_2 - C_3 = \lambda - \sqrt{\lambda^2 - 2\kappa\omega^2} > 0$$

显然成立, 故轨迹为椭圆, 与直角坐标中结论一致。

3.(i) 新的广义动量:

$$\begin{aligned}
 p_r &= -\frac{\partial S}{\partial r} = -\sqrt{2m\lambda - \frac{2m\kappa}{r^2} - m^2\omega^2 r^2} \\
 p_\theta &= -\frac{\partial S}{\partial \theta} = -\sqrt{2m\kappa}
 \end{aligned}$$

绝热不变量可以用留数定理积出:

$$\begin{aligned}
 I_r &= \frac{1}{2\pi} \oint p_r dr = \frac{1}{2\pi} \int_0^{2\pi} p_r \frac{dr}{d\theta} d\theta = \frac{\lambda}{\omega} + \sqrt{2m\kappa} \\
 I_\theta &= \frac{1}{2\pi} \oint p_\theta d\theta = \frac{1}{2\pi} \int_0^{2\pi} (-\sqrt{2m\kappa}) d\theta = -\sqrt{2m\kappa}
 \end{aligned}$$

则:

$$H = \frac{1}{2m} \left[p_r^2 + \frac{p_\theta^2}{r^2} \right] + \frac{1}{2} m \omega^2 r^2$$

$$\begin{aligned}
&= \frac{1}{2m} \left[\left(2m\lambda - \frac{2m\kappa}{r^2} - m^2\omega^2 r^2 \right) + \frac{2m\kappa}{r^2} \right] + \frac{1}{2} m\omega^2 r^2 \\
&= \lambda \\
&= \omega(I_r + I_\theta)
\end{aligned}$$